

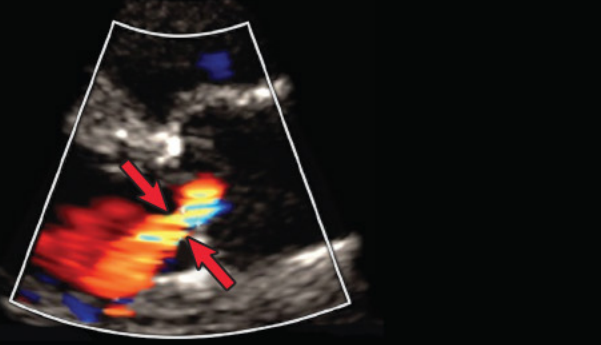
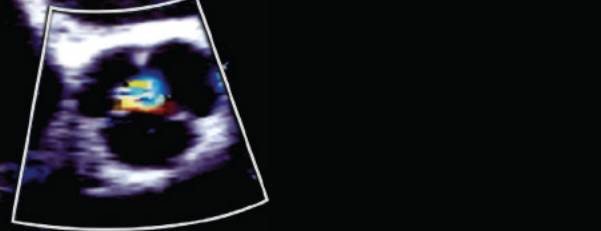
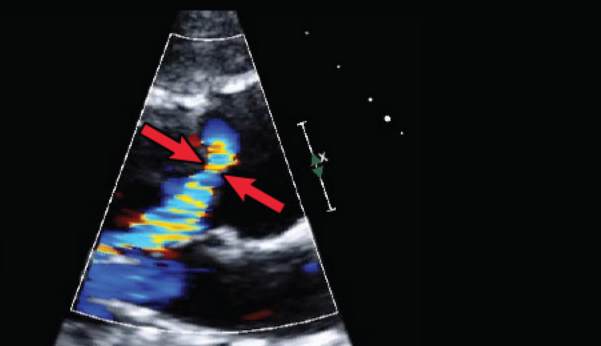
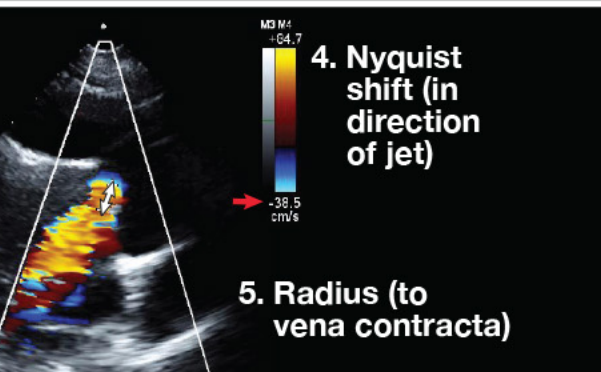
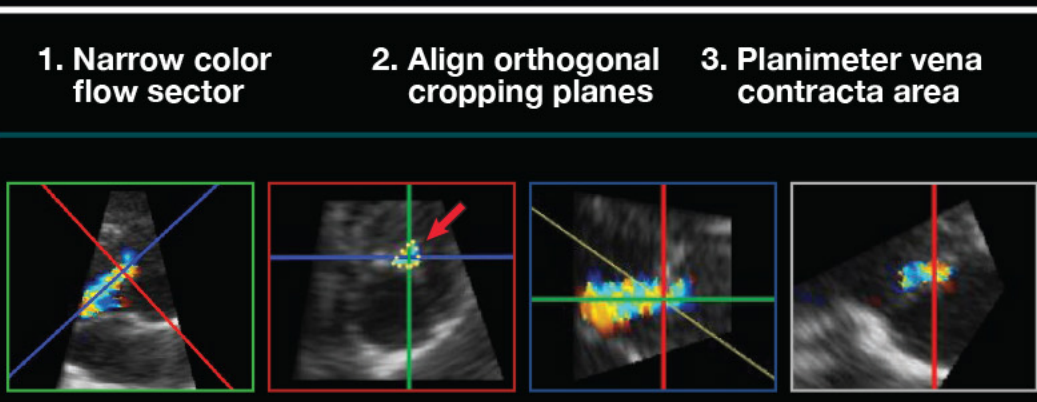


ASE

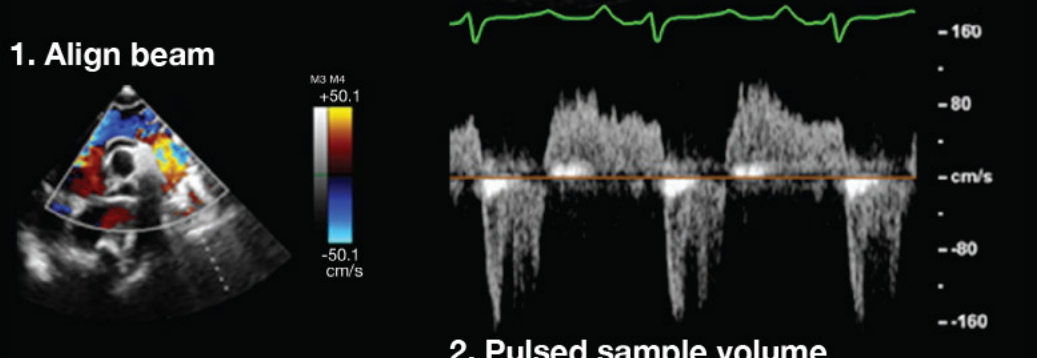
American Society of
Echocardiography

Evaluation of Aortic Regurgitation

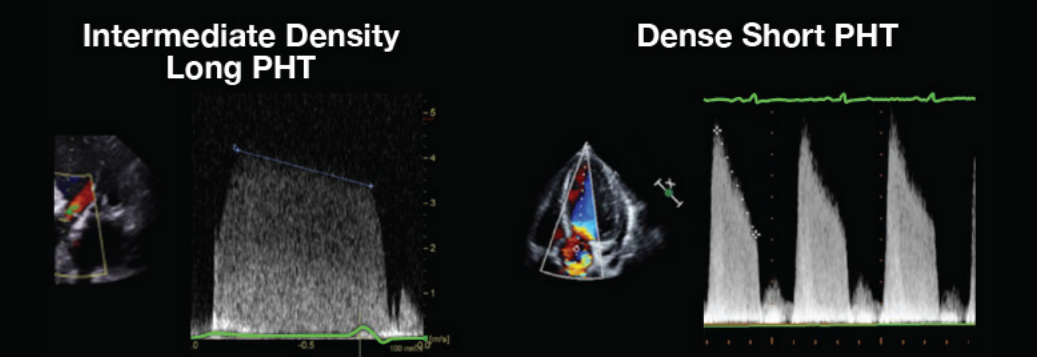
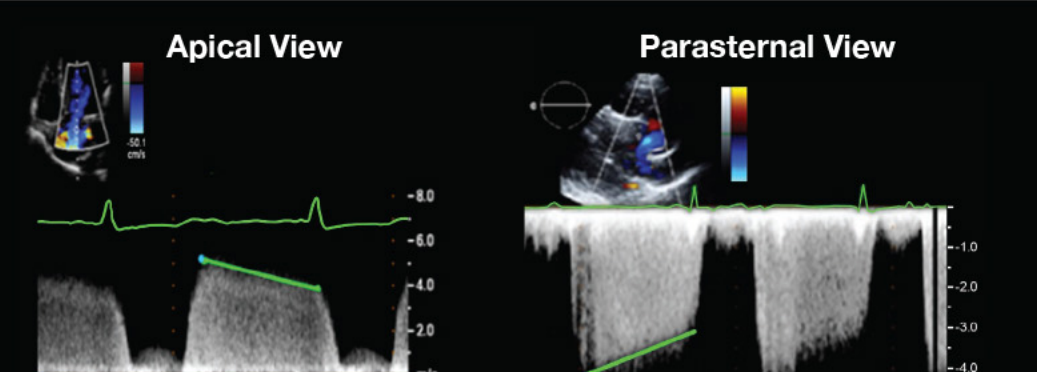
A. Color Flow Doppler (2D and 3D)

Jet Width/LVOT Diameter 1. Long-axis, zoomed view 2. Align jet to optimize VC imaging (may be different from PISA) 3. Measure jet (red arrows) in LVOT within 1 cm of VC 4. Measure LVOT (white arrow)	 1. Long-axis 2. Align jet 3. Measure jet (red arrows) in LVOT within 1 cm of VC 4. Measure LVOT (white arrow)	Advantages: - Simple sensitive screen for AR - Rapid qualitative assessment Disadvantages: - Underestimates AR in eccentric jets - May overestimate AR in central jets as AR jet may expand unpredictably below the orifice - Affected by the size of the LVOT
Jet Area/LVOT Area 1. Short-axis, zoom view 2. Measure in LVOT within 1 cm of the VC	 1. Short-axis 2. Measure in LVOT within 1 cm of the VC	Advantage: - Estimate of regurgitant orifice area Disadvantages: - Direction and shape of jet may overestimate or underestimate jet area
Vena Contracta 1. Long-axis, zoomed view 2. Align jet to optimize VC imaging (may be different from PISA) 3. Measure the narrowest jet diameter at or just apical to the valve	 1. Long-axis 2. Align jet 3. Measure the narrowest jet diameter at or just apical to the valve	Advantages: - Surrogate for regurgitant orifice size - May be used in eccentric jets - Independent of flow rate and driving pressure - Less dependent on technical factors - Good at identifying mild or severe AR Disadvantages: - Presence of multiple jets or bicuspid valves - Convergence zone needs to be visualized - The direction of the jet will influence its appearance
Proximal Flow Convergence 1. Align direction of flow with insonation beam 2. Zoomed view 3. Variance off 4. Change baseline of Nyquist limit (in direction of jet) 5. Measure radius (white arrow in image) from point of color aliasing to vena contracta	 1. Align beam 2. Zoom 3. Variance off 4. Nyquist shift (in direction of jet) 5. Radius (to vena contracta)	Advantage: - Rapid qualitative assessment Disadvantages: - Multiple jets - Constrained jet (aortic wall) - Non-hemispheric shape - Timing in early diastole
3D Vena Contracta 1. Color flow sector should be narrow 2. Align orthogonal cropping planes along the axis of the jet 3. Choose a mid-diastolic cycle 4. Non-coaxial jets or aliased flow may appear "laminar" but still represent regurgitant flow	 1. Narrow color flow sector 2. Align orthogonal cropping planes 3. Planimeter vena contracta area	Advantage: - Multiple jets of differing directions may be measured Disadvantage: - Dynamic jets may be over- or underestimated

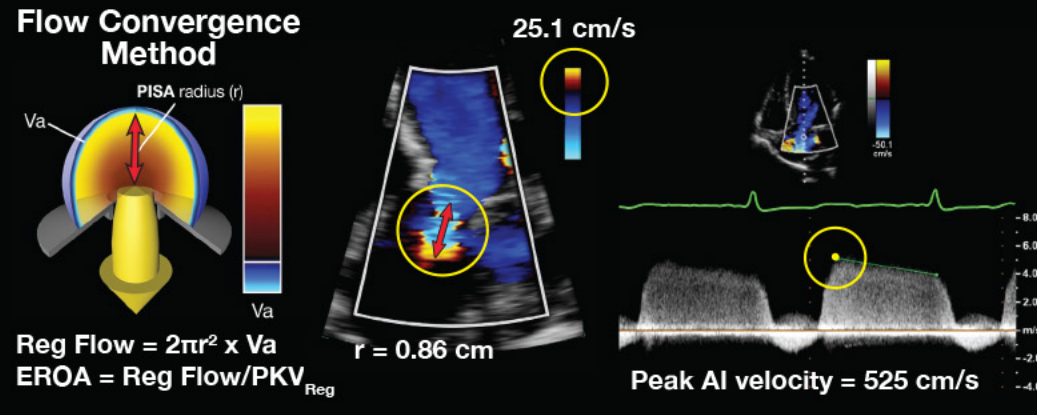
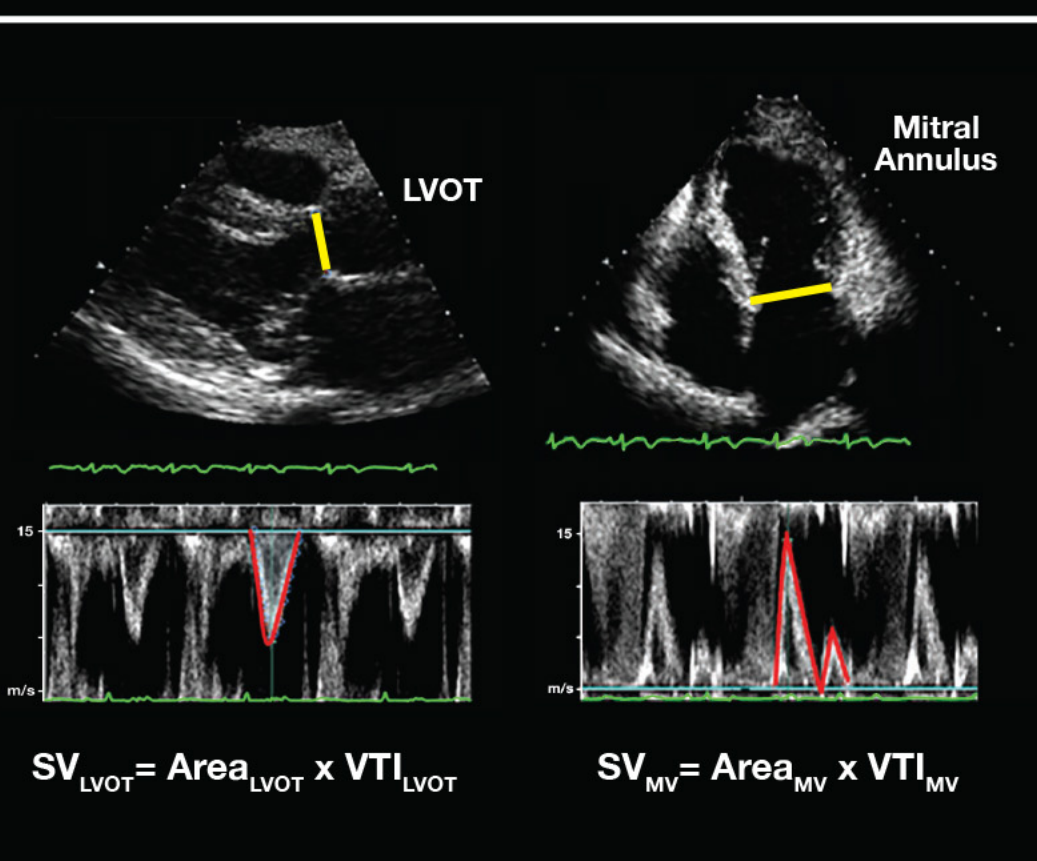
B. Pulsed Wave Doppler

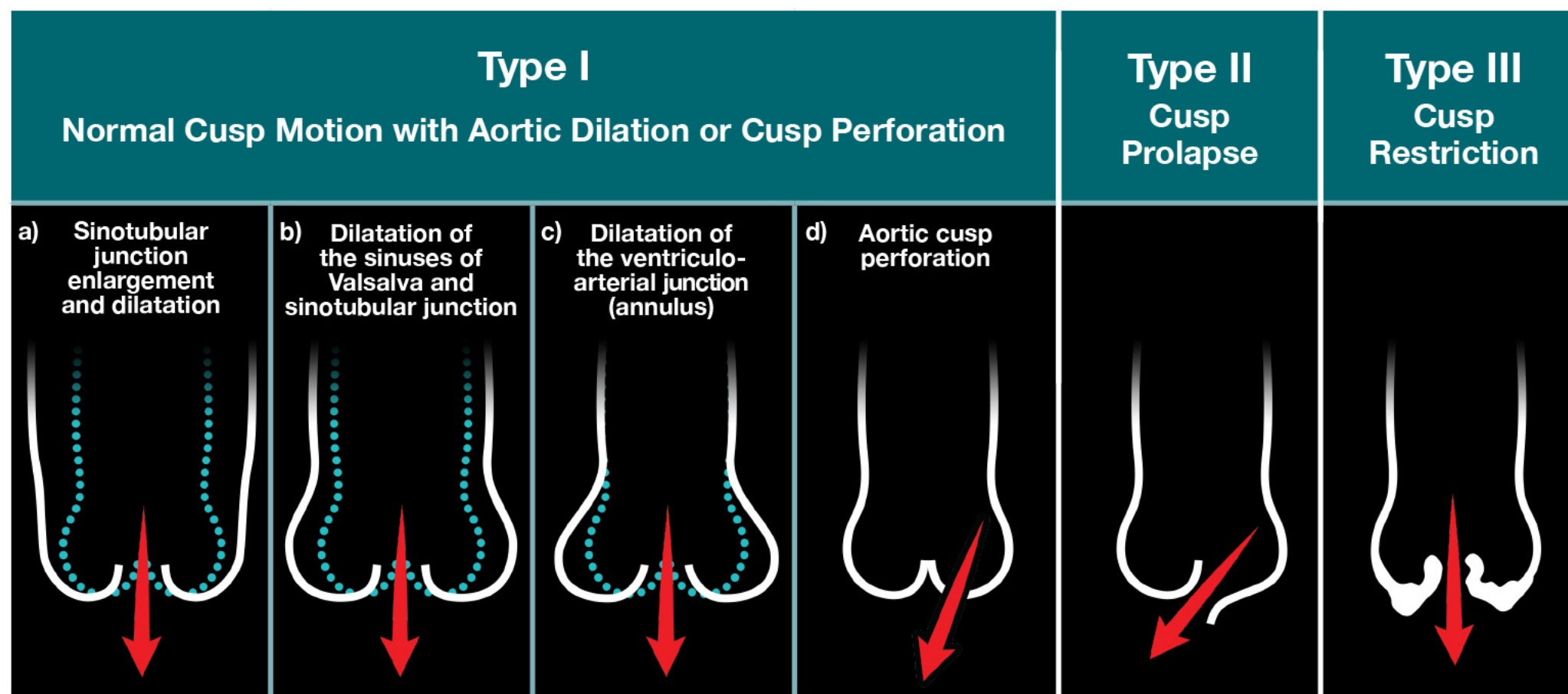
Holodiastolic Flow Reversal in Proximal Descending Aorta 1. Align insonation beam with the flow 2. Pulsed sample volume in the proximal descending or abdominal aorta	 1. Align beam 2. Pulsed sample volume	Advantages: - Simple supportive sign of severe AR - More specific sign if seen in abdominal aorta Disadvantages: - Depends on compliance of the aorta; less reliable in older patients - Brief velocity reversal is normal - May be seen in other conditions - May not be holosystolic in acute AR
--	---	--

C. Continuous Wave Doppler

Density of Regurgitant Jet 1. Align insonation beam with the flow 2. Adjust overall gain	 Intermediate Density Long PHT Dense Short PHT	Advantages: - Simple - Faint or incomplete jet is compatible with mild or trace AR Disadvantages: - Qualitative - Perfectly central jets may appear denser than eccentric jets of higher severity - Overlap between moderate and severe AR
Jet Deceleration Rate (Pressure Half-time) 1. Align insonation beam with the flow 2. Usually best from apical windows 3. In eccentric jets, may be best from parasternal window, helped by color Doppler	 Apical View Parasternal View	Advantages: - Simple - Specific sign of pressure relation between Ao and LV Disadvantages: - Qualitative - Poor alignment of Doppler beam may result in lower PHT - Affected by changes that modify LV-Ao pressure gradient (if short, implies significant AR or high LV filling pressure)

D. Quantitative Doppler: EROA, Regurgitation Volume and Fraction

Flow Convergence Method (PISA): 1. Align insonation beam with the flow 2. Lower the color Doppler baseline in the direction of the jet 3. Look for the hemispheric shape to guide the best lower Nyquist limit 4. CW Doppler of regurgitant jet for peak velocity and VTI	 Flow Convergence Method Reg Flow = $2\pi r^2 \times V_a$ EROA = Reg Flow/PKV _{ring} Peak AI velocity = 525 cm/s	Advantage: - Rapid quantitative assessment of lesion severity (EROA) and volume overload (RVol) Disadvantages: - Feasibility is limited by aortic valve calcifications - Not valid for multiple jets, less accurate in eccentric jets - Small errors in radius measurement can lead to substantial errors in EROA
Stroke Volume Method: Regurgitant Volume= SV_{LVOT} - SV_{MV} 1. LVOT systolic diameter and pulsed Doppler sample volume from different views but at same anatomic level (represents total stroke volume) 2. Mitral mid-diastolic annulus and pulsed Doppler at the same annulus from apical view (represents forward stroke volume) 3. Total LV stroke volume can also be measured by the difference between LV end-diastolic volume and end-systolic volume (best by 3D)	 LVOT Mitral Annulus $SV_{LVOT} = Area_{LVOT} \times VTI_{LVOT}$ $SV_{MV} = Area_{MV} \times VTI_{MV}$	Advantages: - Quantitative, valid with multiple jets, eccentric jets - Provides both lesion severity (EROA, RF) and volume overload (RVol) Disadvantages: - Difficulties measuring mitral annulus diameter, in setting of MR, pulmonary stroke volume used for forward stroke volume - Cumbersome, needs training - Small errors in diameter measurement can lead to substantial errors



Grading the Severity of Chronic AR by Echocardiography¹

Parameters	Mild	Moderate	Severe
Structural Parameters			
Aortic leaflets	Normal or abnormal	Normal or abnormal	Abnormal/flail, or wide coaptation defect
LV size	Normal ²	Normal or dilated	Usually dilated ³
Qualitative Doppler			
Jet width in LVOT, color flow	Small in central jets	Intermediate	Large in central jets; variable in eccentric jets
Flow convergence, color flow	None or very small	Intermediate	Large
Jet density, CW	Incomplete or faint	Dense	Dense
Jet deceleration rate, CW (PHT, msec) ⁴	Incomplete or faint, Slow >500	Medium 500-200	Steep <200
Diastolic flow reversal in descending aorta, PW	Brief, early diastolic reversal	Intermediate	Prominent holodiastolic reversal
Semiquantitative ⁵			
VCW (cm)	<0.3	0.3-0.6	>0.6
Jet width/LVOT width, central jets (%)	<25	25-45	≥65
Jet CSA/LVOT CSA, central jets (%)	<5	5-20	≥60
Quantitative parameters ⁵			
RVol (mL/beat)	<30	30-44	≥60
RF	<30%	30-39%	≥50%
EROA (cm²)	<0.10	0.10-0.19	≥0.30

PHT, Pressure half-time; PW, pulsed wave Doppler.
Color Doppler usually performed at Nyquist limit of 50-70 cm/sec.
1. Bolded signs are considered specific for their AR grade. All parameters have limitations, and an integrated approach must be used that weighs the strength of each echocardiographic measurement. All signs and measures should be interpreted in an individualized manner that accounts for body size, sex, and all other patient characteristics.
2. Unless there are other reasons for LV dilation.
3. Specific in normal LV function, in absence of causes of volume overload. Exception: acute AR, in which chambers have not had time to dilate.
4. PHT is shortened with increasing LV diastolic pressure and may be lengthened in chronic adaption to severe AR.
5. Quantitative parameters can subclassify the moderate regurgitation group.

